

Chapter 02: First-Order Differential Equations & Modeling

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Applied Ordinary Differential Equations (ENGR 213)

Textbook:

Advanced Engineering Mathematics

(7th Edition) by Dennis G. Zill — Chapter 2 (§2.1, §2.2, §2.3, §2.4, §2.5, §2.7, §2.8)

1. Executive Summary & Road Map

Chapter 2 forms the analytical and practical core of first-order ordinary differential equations. In this chapter, we master the complete toolkit for analyzing, solving, and interpreting first-order equations:

First-Order Differential Equations (§2.1 - §2.8)

- └ Qualitative Analysis (§2.1)
 - └ Direction fields & lineal elements
 - └ Autonomous ODEs, phase line portraits & stability (attractor, repeller, semi-stable)
- └ Analytical Methods (§2.2 - §2.5)
 - └ §2.2 Separation of Variables: $g(x)dx = h(y)dy$ & lost singular solutions
 - └ §2.3 First-Order Linear: $y' + P(x)y = f(x)$ via Integrating Factor $\mu(x) = \exp(\int P dx)$
 - └ §2.4 Exact Equations: $M dx + N dy = 0$ ($M_y = N_x$) & integrating multipliers
 - └ §2.5 Substitutions:
 - └ Homogeneous equations: $y = u \cdot x$
 - └ Bernoulli equations: $w = y^{(1-n)}$
 - └ Linear composition: $u = Ax + By + C$
- └ Physical Modeling (§2.7 - §2.8)
 - └ §2.7 Linear Models: growth/decay, Newton cooling, mixture tanks (equal/unequal rates), LR/RC circuits
 - └ §2.8 Nonlinear Models: logistic population dynamics, carrying capacity, Torricelli tank draining

2. Qualitative Analysis: Direction Fields & Stability (§2.1)

2.1 Direction Fields and Flow Trajectories

When an analytical formula for the solution of $\frac{dy}{dx} = f(x, y)$ cannot be obtained, or when global behavior must be visualized immediately, we employ **Direction Fields** (also known as slope fields).

- Lineal Elements:** At each point (x, y) , the value $f(x, y)$ represents the slope of the tangent line. A short line segment with slope $m = f(x, y)$ centered at (x, y) is a **lineal element**.
- Isoclines:** Curves along which the slope is constant ($f(x, y) = c$). Connecting points of identical slope assists in hand-sketching direction fields.
- Integral Curves:** Any solution $y = \phi(x)$ must pass through the direction field such that its tangent line at every point coincides with the lineal element.

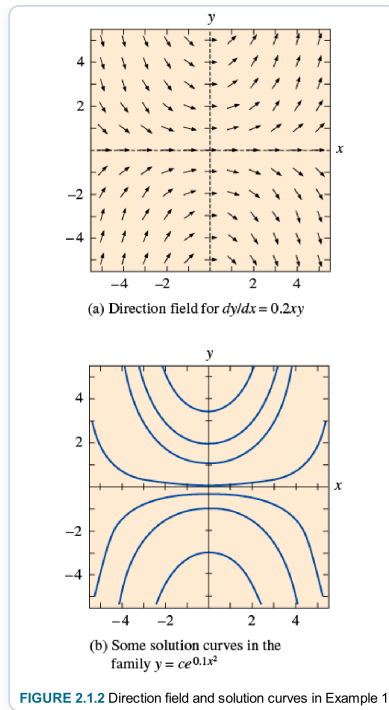


Figure 2.1: Direction field and family of integral solution curves for $\frac{dy}{dx} = 0.2xy$ — extracted directly from Official Textbook (7th Ed., Chapter 2, Fig. 2.1.2).

In-Depth Pedagogical Breakdown:

- **Symmetries and Zeros (Fig. 2.1.2a):**

- Along the axes ($x = 0$ or $y = 0$), the slope is strictly zero ($0.2(0) = 0$). All lineal elements on the axes are strictly horizontal.
- In Quadrants I ($x > 0, y > 0$) and III ($x < 0, y < 0$), the product $xy > 0$, so all slopes are positive.
- In Quadrants II ($x < 0, y > 0$) and IV ($x > 0, y < 0$), the product $xy < 0$, so all slopes are negative.

- **Integral Solution Curves (Fig. 2.1.2b):**

- Solving $\frac{dy}{y} = 0.2x dx \implies \ln|y| = 0.1x^2 + C_1 \implies y(x) = ce^{0.1x^2}$.
- For $c > 0$, the curves form upward-opening exponential profiles.
- The line $y \equiv 0$ ($c = 0$) serves as an equilibrium barrier that solution curves never cross.

2.2 Autonomous First-Order ODEs & Phase Line Analysis

An ODE is **autonomous** if the independent variable x (or time t) does not appear explicitly:

$$\frac{dy}{dx} = f(y)$$

A. Critical Points & Equilibrium Solutions

The zeros of the function $f(y)$ are called **critical points**, **equilibrium points**, or **stationary points**. If $f(c) = 0$, then the constant function:

$$y(x) \equiv c$$

is a solution of the ODE, called an **equilibrium solution**. On the direction field, equilibrium solutions are horizontal straight lines.

B. The One-Dimensional Phase Line

The critical points divide the y -axis into distinct open intervals. Because $f(y)$ is continuous, it cannot change sign on any interval between adjacent critical points:

- If $f(y) > 0$ on (c_1, c_2) , then $\frac{dy}{dx} > 0$, meaning every solution curve in this region is strictly **increasing** (represented by an **upward arrow** $\uparrow$ on the phase line).
- If $f(y) < 0$ on (c_1, c_2) , then $\frac{dy}{dx} < 0$, meaning every solution curve in this region is strictly **decreasing** (represented by a **downward arrow** $\downarrow$ on the phase line).

C. Classification of Critical Points

1. Asymptotically Stable (Attractor / Sink):

- Arrows on both sides point toward c ($\downarrow$ from above, $\uparrow$ from below).
- Any solution starting near c converges to c as $x \rightarrow \infty$: $\lim_{x \rightarrow \infty} y(x) = c$.
- **Derivative Test:** If $f'(c) < 0$, then c is asymptotically stable.

2. Unstable (Repeller / Source):

- Arrows on both sides point away from c ($\uparrow$ above, $\downarrow$ below).
- Solutions diverge away from c as $x \rightarrow \infty$.
- **Derivative Test:** If $f'(c) > 0$, then c is unstable.

3. Semi-Stable (Shunt):

- Arrows on both sides point in the same direction ($\uparrow$ on both sides, or $\downarrow$ on both sides).
- Solutions approach c from one side but diverge from the other. Occurs when $f(y)$ has a root of even multiplicity (e.g., $f(y) = (y - c)^2$).
- **Derivative Test:** If $f'(c) = 0$ and $f''(c) \neq 0$, c is semi-stable.

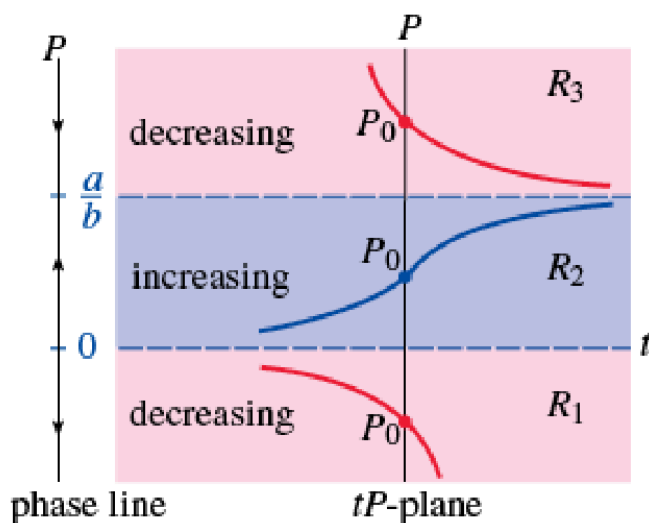


FIGURE 2.1.6 Phase portrait and solution curves in each of the three subregions in Example 4

Figure 2.2: Autonomous phase line mapped to solution trajectories in the tP -plane across regions R_1, R_2, R_3 — from Textbook 7th Ed. (Fig. 2.1.6).

D. Translation Property of Autonomous ODEs

If $y(x)$ is a solution of an autonomous ODE $\frac{dy}{dx} = f(y)$, then $y(x - x_0)$ is also a solution for any constant x_0 .
Geometrical Meaning

: Every solution curve in a given region between critical points is simply a horizontal translation of any other solution curve in that region.

3. Analytical Solution Engine (§2.2 - §2.5)

3.1 Separation of Variables (§2.2)

An equation is separable if it can be factored into a product of a function of x and a function of y :

$$\frac{dy}{dx} = g(x)h(y)$$

Step-by-Step Algorithm:

1. **Identify Critical Points:** Solve $h(y) = 0$. Each real root $y = r$ gives an immediate constant solution $y(x) \equiv r$.
2. **Separate:** Assuming $h(y) \neq 0$, divide by $h(y)$ and multiply by dx :

$$\frac{1}{h(y)} dy = g(x) dx$$

3. **Integrate:**

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C$$

4. **Isolate y (if possible):** Solve explicitly for $y = \phi(x, C)$.
5. **Check for Lost (Singular) Solutions:** Compare the constant solutions $y \equiv r$ found in Step 1 with the family obtained in Step 4. If $y \equiv r$ cannot be obtained by any choice of C , it is a **singular solution** and must be listed separately.

3.2 First-Order Linear Equations (§2.3)

A first-order linear ODE can always be written in **standard form**:

$$\frac{dy}{dx} + P(x)y = f(x)$$

Theoretical Derivation of the Integrating Factor:

We seek a multiplying function $\mu(x)$ such that the left-hand side transforms into the derivative of a single product:

$$\mu(x) \left[\frac{dy}{dx} + P(x)y \right] = \frac{d}{dx} [\mu(x)y] = \mu(x) \frac{dy}{dx} + \mu'(x)y$$

Equating coefficients of y :

$$\mu(x)P(x) = \mu'(x) \implies \frac{1}{\mu} \frac{d\mu}{dx} = P(x) \implies \ln |\mu| = \int P(x) dx \implies \mu(x) = e^{\int P(x) dx}$$

Complete Master Solution Formula:

1. Put the ODE in standard form (divide by leading coefficient $a_1(x)$ if $a_1(x) \neq 1$):

$$\frac{dy}{dx} + P(x)y = f(x)$$

2. Calculate the integrating factor:

$$\mu(x) = e^{\int P(x)dx}$$

3. Multiply the entire standard equation by $\mu(x)$:

$$\frac{d}{dx}[\mu(x)y] = \mu(x)f(x)$$

4. Integrate both sides with respect to x :

$$\mu(x)y = \int \mu(x)f(x)dx + C$$

5. Divide by $\mu(x)$ to obtain the explicit general solution:

$$y(x) = \frac{1}{\mu(x)} \int \mu(x)f(x)dx + \frac{C}{\mu(x)}$$

Theorem 2.3.1 (Continuity & Interval of Definition):

If $P(x)$ and $f(x)$ are continuous on an open interval I containing x_0 , then there exists a unique solution to the IVP $y' + P(x)y = f(x)$; $y(x_0) = y_0$, and this solution is **continuous on the entire interval I** .

Critical Insight

: Unlike nonlinear equations, the domain of validity of a linear ODE's solution is completely known in advance simply by finding the discontinuities of $P(x)$ and $f(x)$!

3.3 Exact Equations & Integrating Multipliers (§2.4)

A differential expression $M(x, y)dx + N(x, y)dy$ is an **exact differential** in a region R of the xy -plane if it corresponds to the total differential df of some multivariable potential function $f(x, y)$:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = M(x, y)dx + N(x, y)dy$$

Theorem 2.4.1 (Test for Exactness):

Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first partial derivatives in a simply connected region R . Then:

$$M(x, y)dx + N(x, y)dy = 0 \text{ is EXACT} \iff \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Exhaustive Step-by-Step Solving Algorithm:

- **Step 1:** Check exactness: compute $M_y = \frac{\partial M}{\partial y}$ and $N_x = \frac{\partial N}{\partial x}$. If $M_y = N_x$, proceed.

- **Step 2:** Since $\frac{\partial f}{\partial x} = M(x, y)$, integrate M with respect to x (treating y as a constant):

$$f(x, y) = \int M(x, y) dx + g(y)$$

where $g(y)$ is the arbitrary function of integration.

- **Step 3:** Differentiate $f(x, y)$ with respect to y :

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \left[\int M(x, y) dx \right] + g'(y)$$

- **Step 4:** Set this partial derivative equal to $N(x, y)$ and solve for $g'(y)$:

$$\frac{\partial}{\partial y} \left[\int M(x, y) dx \right] + g'(y) = N(x, y) \implies g'(y) = N(x, y) - \frac{\partial}{\partial y} \left[\int M(x, y) dx \right]$$

(Verification checkpoint: all x variables MUST cancel out. If x remains in $g'(y)$, an error occurred!)

- **Step 5:** Integrate $g'(y)$ with respect to y to obtain $g(y)$.
- **Step 6:** State the final implicit solution:

$$f(x, y) = C$$

Integrating Multipliers for Non-Exact Equations:

If $M_y \neq N_x$, multiply by an integrating factor $\mu(x, y)$ such that $(\mu M)_y = (\mu N)_x$:

1. **Case 1: Factor of x alone:** If $\frac{M_y - N_x}{N} = p(x)$ (depends only on x), then:

$$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}$$

2. **Case 2: Factor of y alone:** If $\frac{N_x - M_y}{M} = q(y)$ (depends only on y), then:

$$\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}$$

3.4 Solutions by Substitution (§2.5)

Substitutions Toolkit (§2.5)

- └ 1. Homogeneous Equations: $M(tx, ty) = t^n M(x, y)$ & $N(tx, ty) = t^n N(x, y)$
 - └ Substitute $y = u \cdot x \implies dy = u dx + x du$ (Separable in u and x)
- └ 2. Bernoulli Equations: $y' + P(x)y = Q(x)y^n$
 - └ Multiply by $y^{-(n)}$, let $w = y^{(1-n)} \implies$ Linear in w !
- └ 3. Linear Composition: $y' = f(Ax + By + C)$
 - └ Let $u = Ax + By + C \implies du/dx = A + B \cdot y'$ (Separable in u and x)

A. Homogeneous Equations

A function $f(x, y)$ is homogeneous of degree n if $f(tx, ty) = t^n f(x, y)$. If $M(x, y)$ and $N(x, y)$ are both homogeneous of the same degree n , the ODE $Mdx + Ndy = 0$ can be written as $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$.

- **Substitution:** Let $y = ux \implies \frac{dy}{dx} = u + x \frac{du}{dx}$.

- **Separable Form:**

$$u + x \frac{du}{dx} = F(u) \implies x \frac{du}{dx} = F(u) - u \implies \frac{du}{F(u) - u} = \frac{dx}{x}$$

- *Alternative*

: If $M(x, y)$ is simpler than $N(x, y)$, substitute $x = vy \implies dx = vdy + ydv$.

B. Bernoulli Equations

An ODE of the form:

$$\frac{dy}{dx} + P(x)y = f(x)y^n, \quad n \neq 0, 1$$

is called a **Bernoulli Equation**.

- **Baby-Step Linearization:**

1. Divide through by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = f(x)$$

2. Substitute $w = y^{1-n}$. By the chain rule:

$$\frac{dw}{dx} = (1-n)y^{-n} \frac{dy}{dx} \implies y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dw}{dx}$$

3. Substitute into the equation:

$$\frac{1}{1-n} \frac{dw}{dx} + P(x)w = f(x)$$

4. Multiply by $(1-n)$ to achieve standard linear form:

$$\frac{dw}{dx} + (1-n)P(x)w = (1-n)f(x)$$

5. Solve for $w(x)$ using an integrating factor, then back-substitute $y = w^{1/(1-n)}$.

C. Reduction to Separation of Variables

An ODE of the form $\frac{dy}{dx} = f(Ax + By + C)$ with $B \neq 0$:

1. Substitute $u = Ax + By + C$.

2. Differentiate with respect to x : $\frac{du}{dx} = A + B \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{B} \left(\frac{du}{dx} - A \right)$.

3. Substitute: $\frac{1}{B} \left(\frac{du}{dx} - A \right) = f(u) \implies \frac{du}{dx} = A + Bf(u)$.

4. Separate: $\frac{du}{A+Bf(u)} = dx$.

4. Applied Physical Modeling Engine (§2.7 - §2.8)

4.1 Industrial Mixture Tanks (Equal vs. Unequal Flow Rates)

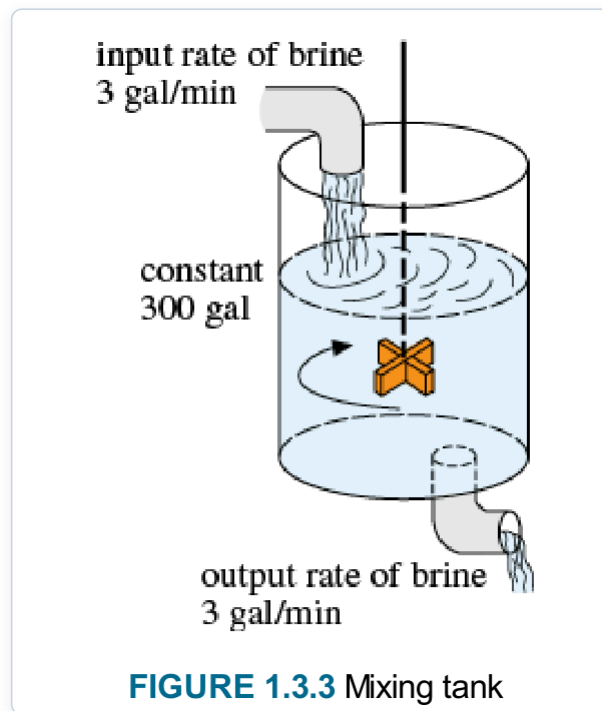


Figure 2.3: Mixing tank schematic showing fluid inflow, impeller agitation, and bottom drainage — from Textbook 7th Ed. (Fig. 1.3.3).

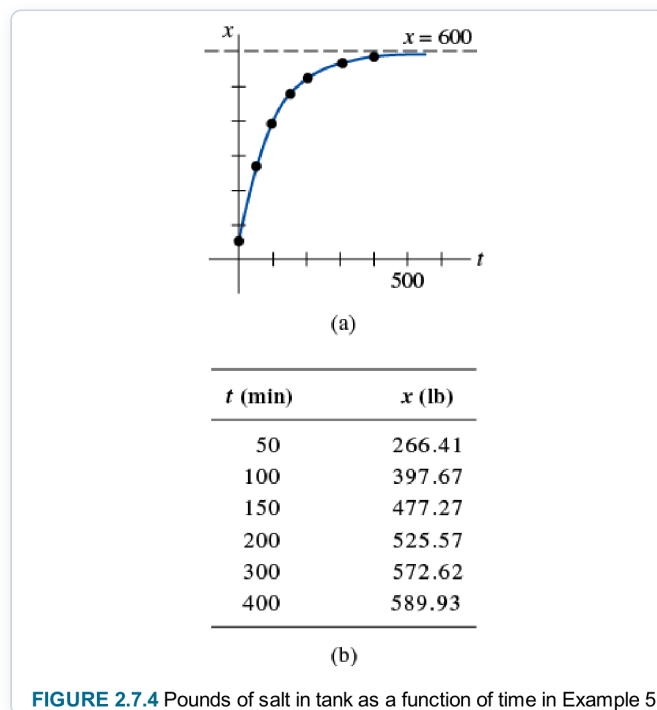


Figure 2.4: Salt accumulation curve $x(t)$ approaching horizontal asymptote $x = 600$ lb — from Textbook 7th Ed. (Fig. 2.7.4).

In-Depth Mathematical Modeling:

1. Governing Equation:

$$\frac{dA}{dt} + \left(\frac{r_{\text{out}}}{V_0 + (r_{\text{in}} - r_{\text{out}})t} \right) A = r_{\text{in}} c_{\text{in}}$$

2. Case Study (Equal Flow Rates $r_{\text{in}} = r_{\text{out}} = 3$ gal/min):

- Tank volume $V(t) = 300$ gal, initial salt $A(0) = 50$ lb, input concentration $c_{\text{in}} = 2$ lb/gal.
- Input rate: $R_{\text{in}} = (3 \text{ gal/min})(2 \text{ lb/gal}) = 6 \text{ lb/min}$.
- Output rate: $R_{\text{out}} = (3 \text{ gal/min}) \left(\frac{A}{300} \text{ lb/gal} \right) = \frac{1}{100} A \text{ lb/min}$.
- Standard Form: $\frac{dA}{dt} + \frac{1}{100} A = 6$.
- Integrating Factor: $\mu(t) = e^{\int 0.01 dt} = e^{0.01t}$.
- Integration: $\frac{d}{dt}[e^{0.01t} A] = 6e^{0.01t} \implies e^{0.01t} A = 600e^{0.01t} + C \implies A(t) = 600 + Ce^{-0.01t}$.
- Apply $A(0) = 50 \implies 50 = 600 + C \implies C = -550$.
- Analytical Solution:

$$A(t) = 600 - 550e^{-t/100} \text{ lb}$$

- As $t \rightarrow \infty$, $A(t) \rightarrow 600$ lb, which equals $V \times c_{\text{in}} = 300 \times 2 = 600$ lb (asymptote in Fig. 2.7.4).

4.2 Nonlinear Population Dynamics: The Logistic Model (§2.8)

In 1840, Pierre Franois Verhulst modified the Malthusian law by adding an environment-limiting competition term $-bP^2$:

$$\frac{dP}{dt} = P(a - bP), \quad a, b > 0$$

Writing in terms of intrinsic growth rate $r = a$ and carrying capacity $K = a/b$:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right)$$

$$\begin{aligned} \frac{d^2P}{dt^2} &= P \left(-b \frac{dP}{dt} \right) + (a - bP) \frac{dP}{dt} = \frac{dP}{dt} (a - 2bP) \\ &= P(a - bP)(a - 2bP) \\ &= 2b^2P \left(P - \frac{a}{b} \right) \left(P - \frac{a}{2b} \right). \end{aligned}$$

Figure 2.5: Family of logistic curves showing sigmoidal growth and asymptotic convergence to carrying capacity K — from Textbook 7th Ed. (Fig. 2.8.2).

Exhaustive Step-by-Step Derivation of the Logistic Formula:

1. Separate Variables:

$$\frac{dP}{P(a - bP)} = dt$$

2. Partial Fraction Decomposition:

$$\frac{1}{P(a-bP)} = \frac{A}{P} + \frac{B}{a-bP} \implies 1 = A(a-bP) + BP$$

- Setting $P = 0 \implies 1 = A(a) \implies A = \frac{1}{a}$.
- Setting $P = \frac{a}{b} \implies 1 = B\left(\frac{a}{b}\right) \implies B = \frac{b}{a}$.

$$\frac{1}{a} \left(\frac{1}{P} + \frac{b}{a-bP} \right) dP = dt \implies \left(\frac{1}{P} + \frac{b}{a-bP} \right) dP = a, dt$$

3. Integrate Both Sides:

$$\ln|P| - \ln|a-bP| = at + C_1 \implies \ln \left| \frac{P}{a-bP} \right| = at + C_1 \implies \frac{P}{a-bP} = Ce^{at}$$

4. Solve for $P(t)$:

$$P = Ce^{at}(a-bP) = aCe^{at} - bCe^{at}P \implies P(1+bCe^{at}) = aCe^{at}$$

$$P(t) = \frac{aCe^{at}}{1+bCe^{at}} = \frac{aC}{e^{-at}+bC}$$

5. Apply Initial Condition $P(0) = P_0$:

$$\frac{P_0}{a-bP_0} = C$$

Substituting C and simplifying yields the **Canonical Logistic Formula**:

$$P(t) = \frac{aP_0}{bP_0 + (a-bP_0)e^{-at}} = \frac{KP_0}{P_0 + (K-P_0)e^{-rt}}$$

Important Qualitative Features:

- **Asymptote:** For any initial population $P_0 > 0$:

$$\lim_{t \rightarrow \infty} P(t) = \frac{KP_0}{P_0 + 0} = K$$

- **Inflection Point:** Differentiating $\frac{dP}{dt} = aP - bP^2$:

$$\frac{d^2P}{dt^2} = (a-2bP)\frac{dP}{dt} = (a-2bP)P(a-bP)$$

The inflection point occurs at $a-2bP = 0 \implies P = \frac{a}{2b} = \frac{K}{2}$. The growth rate $\frac{dP}{dt}$ is at its **maximum** when the population reaches exactly **half its carrying capacity**!

5. Fully Worked Master Archetypes

Archetype 5.1: Linear First-Order ODE with Trigonometric Integrating Factor

Problem: Solve the initial-value problem:

$$x \frac{dy}{dx} + 2y = \frac{\sin x}{x}, \quad y(\pi) = 1, \quad x > 0$$

Step-by-Step Solution:

- **Step 1: Convert to Standard Form:** Divide the entire equation by x :

$$\frac{dy}{dx} + \frac{2}{x}y = \frac{\sin x}{x^2}$$

Here $P(x) = \frac{2}{x}$ and $f(x) = \frac{\sin x}{x^2}$. Both are continuous on $(0, \infty)$.

- **Step 2: Calculate the Integrating Factor:**

$$\mu(x) = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln(x^2)} = x^2$$

- **Step 3: Multiply Standard Form by $\mu(x)$:**

$$x^2 \frac{dy}{dx} + 2xy = \sin x \implies \frac{d}{dx}[x^2 y] = \sin x$$

- **Step 4: Integrate Both Sides:**

$$x^2 y = \int \sin x, dx = -\cos x + C$$

- **Step 5: Isolate $y(x)$:**

$$y(x) = -\frac{\cos x}{x^2} + \frac{C}{x^2}$$

- **Step 6: Apply Initial Condition $y(\pi) = 1$:**

$$1 = -\frac{\cos(\pi)}{\pi^2} + \frac{C}{\pi^2} = -\frac{-1}{\pi^2} + \frac{C}{\pi^2} = \frac{1+C}{\pi^2}$$
$$\pi^2 = 1 + C \implies C = \pi^2 - 1$$

- **Final Solution:**

$$y(x) = \frac{\pi^2 - 1 - \cos x}{x^2}, \quad x \in (0, \infty)$$

Archetype 5.2: Exact ODE Requiring an Integrating Multiplier $\mu(y)$

Problem: Solve the differential equation:

$$(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$$

Step-by-Step Solution:

- **Step 1: Test for Exactness:**

$$M(x, y) = 2xy^4e^y + 2xy^3 + y \implies \frac{\partial M}{\partial y} = 2x(4y^3e^y + y^4e^y) + 6xy^2 + 1 = 8xy^3e^y + 2xy^4e^y + 6xy^2 + 1$$

$$N(x, y) = x^2y^4e^y - x^2y^2 - 3x \implies \frac{\partial N}{\partial x} = 2xy^4e^y - 2xy^2 - 3$$

$$M_y - N_x = 8xy^3e^y + 8xy^2 + 4 = 4(2xy^3e^y + 2xy^2 + 1)$$

Because $M_y \neq N_x$, the equation is not exact.

- **Step 2: Evaluate Integrating Factor Criteria:** Notice $M(x, y) = y(2xy^3e^y + 2xy^2 + 1)$. Therefore:

$$\frac{N_x - M_y}{M} = \frac{-4(2xy^3e^y + 2xy^2 + 1)}{y(2xy^3e^y + 2xy^2 + 1)} = -\frac{4}{y}$$

This depends solely on y !

- **Step 3: Compute Integrating Factor $\mu(y)$:**

$$\mu(y) = e^{\int -\frac{4}{y} dy} = e^{-4 \ln |y|} = y^{-4} = \frac{1}{y^4}$$

- **Step 4: Multiply the Entire ODE by $\mu(y) = y^{-4}$:**

$$\left(2xe^y + \frac{2x}{y} + \frac{1}{y^3}\right) dx + \left(x^2e^y - \frac{x^2}{y^2} - \frac{3x}{y^4}\right) dy = 0$$

- **Step 5: Re-test Exactness:**

$$M_{\text{new},y} = 2xe^y - \frac{2x}{y^2} - \frac{3}{y^4}$$

$$N_{\text{new},x} = 2xe^y - \frac{2x}{y^2} - \frac{3}{y^4} \quad \checkmark \text{ EXACT!}$$

- **Step 6: Integrate with Respect to x :**

$$f(x, y) = \int \left(2xe^y + \frac{2x}{y} + \frac{1}{y^3}\right) dx = x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} + g(y)$$

- **Step 7: Match with N_{new} :**

$$\frac{\partial f}{\partial y} = x^2e^y - \frac{x^2}{y^2} - \frac{3x}{y^4} + g'(y) = x^2e^y - \frac{x^2}{y^2} - \frac{3x}{y^4}$$

$$g'(y) = 0 \implies g(y) = C_1$$

- **Final General Solution:**

$$x^2e^y + \frac{x^2}{y} + \frac{x}{y^3} = C$$

Archetype 5.3: Bernoulli Equation Transformation

Problem: Solve the initial-value problem:

$$x^2 \frac{dy}{dx} - 2xy = 3y^4, \quad y(1) = \frac{1}{2}$$

Step-by-Step Solution:

- **Step 1: Identify Bernoulli Form and Divide by x^2 :**

$$\frac{dy}{dx} - \frac{2}{x}y = \frac{3}{x^2}y^4$$

Here $P(x) = -\frac{2}{x}$, $Q(x) = \frac{3}{x^2}$, and $n = 4$.

- **Step 2: Divide by y^4 :**

$$y^{-4} \frac{dy}{dx} - \frac{2}{x} y^{-3} = \frac{3}{x^2}$$

- **Step 3: Define the Substitution:**

$$w = y^{1-n} = y^{1-4} = y^{-3}$$

$$\frac{dw}{dx} = -3y^{-4} \frac{dy}{dx} \implies y^{-4} \frac{dy}{dx} = -\frac{1}{3} \frac{dw}{dx}$$

- **Step 4: Substitute into the ODE:**

$$-\frac{1}{3} \frac{dw}{dx} - \frac{2}{x} w = \frac{3}{x^2}$$

Multiply by -3 :

$$\frac{dw}{dx} + \frac{6}{x} w = -\frac{9}{x^2}$$

- **Step 5: Solve the Linear ODE for $w(x)$:**

$$\mu(x) = e^{\int \frac{6}{x} dx} = e^{6 \ln x} = x^6$$

$$\frac{d}{dx}[x^6 w] = x^6 \left(-\frac{9}{x^2} \right) = -9x^4$$

$$x^6 w = \int -9x^4 dx = -\frac{9}{5} x^5 + C$$

$$w(x) = -\frac{9}{5x} + \frac{C}{x^6}$$

- **Step 6: Back-Substitute $w = y^{-3}$:**

$$\frac{1}{y^3} = \frac{C - \frac{9}{5} x^5}{x^6} \implies y(x) = \left(\frac{x^6}{C - \frac{9}{5} x^5} \right)^{1/3}$$

- **Step 7: Apply Initial Condition $y(1) = 1/2$:**

$$y(1) = \frac{1}{2} \implies \frac{1}{(1/2)^3} = 8 = \frac{C - 9/5}{1} \implies C = 8 + \frac{9}{5} = \frac{49}{5}$$

- **Final Explicit Solution:**

$$y(x) = \left(\frac{5x^6}{49 - 9x^5} \right)^{1/3}$$

6. Diagnostic Traps, Common Pitfalls & Exam Warnings

-  **Trap 1: Dropping Negative Signs in Integrating Factors:** For $\frac{dy}{dx} - \frac{3}{x} y = x^2$, the coefficient is $P(x) = -\frac{3}{x}$.

$$\mu(x) = e^{\int -\frac{3}{x} dx} = e^{-3 \ln x} = x^{-3} = \frac{1}{x^3}$$

Never

write x^3 ! Dropping the negative sign completely ruins the product rule on the LHS.

-  **Trap 2: Forgetting to Multiply $f(x)$ by the Integrating Factor:**

$$\frac{d}{dx}[\mu(x)y] = \mu(x)f(x)$$

Students frequently write $\frac{d}{dx}[\mu y] = f(x)$, forgetting to multiply the right-hand side by $\mu(x)$ before integrating.

-  **Trap 3: Variable Volume Mixing Pitfalls:** When $r_{\text{in}} \neq r_{\text{out}}$, the volume is $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$.
Common mistake

: Treating volume as constant V_0 . This turns a variable-coefficient integrating factor $\mu(t) = (V_0 + \Delta r \cdot t)^k$ into an incorrect exponential e^{kt} .

-  **Trap 4: Forgetting the Integrating Constant on Partial Integrals in Exact ODEs:** When integrating $M(x, y)$ with respect to x , the constant of integration is a function of y : $g(y)$. Writing a numerical constant $+C$ prevents solving for $N(x, y)$ and breaks the exact algorithm.